Luca Predieri — Curriculum Vitae

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Embedded Software Engineer at Moog Inc., MSc in Robotics Engineering at Università degli Studi di Genova. My main interests are robotics, low-level software and the studies of engineering related to the development of modern machines with duties in industrial automation.

Previous Employment

Embedded Systems Engineer — **Casella, Italy**
Moog Inc., Full-Time — *September 2023 – Present*

I am presently serving as an embedded software engineer at Moog, a globally renowned leader in the field of automation engineering. My primary responsibilities revolve around software development and conducting comprehensive research, with a particular focus on the interplay between software and hardware, as well as mechanical control. My work focussed on the developmente of the latest products available in the market. The work was focussed mainly of DSP families by Texas Instruments.

- Development of firmware for C2000 DSP based products, such as servo-drives, and PSUs.
- Developing code for peripherals such as SPI, SCI, I2C, DMAs, GPIOs, Timers, UART, DACs, ADCs and PWMs.
- Held on-hand tests on board prototypes, used a wide range of laboratory instruments such as oscilloscopes, multi-meters, DC supplies and others.
- We discuss design choices for hardware on prototypes.
- Developing and analyzing solutions for Cyber Resilience Act (CRA) for our products.
- Moog Italy AI Ambassador for improving our code development with AI tools, such as Codex, Antigravity, Claude and other agent like tools.
- Deepen my personal knowledge on development of electronic boards, in an industrial and operations point of view. In the site operations, quality, R&D and production are all sectors with people working together to achieve goals.

Embedded Software Engineer — **Genova, Italy**
ARMS (Automated Reefer Management System) S.r.l., Freelance — *May 2023 – August 2023*

While working for the thesis, I covered some tasks in a fast-growing start-up in Genova related to the development of cartesian robots and manipulators for the shipping world. I provided consultancy or contributed to the company's tasks on an occasional or ad hoc basis:

- Evaluation of sensors for the low-cost manipulator, I changed sensors at the end-effector.
- Public speaking for presenting the project in local events.
- Evaluation and proposals for improving the projects with robotics-oriented ideas.

- **Software Engineer** — **Genova, Italy**
  Esaote S.p.A., Internship — *February 2023 – September 2023*
  I had the opportunity to work for my master thesis at Esaote, a big company which is one of the leader companies in diagnostic imaging in the world. My main mission in the company was to assist my more experienced colleagues in completing the team's objectives more quickly. Some tasks I covered were:
  - Held discussions and study for Matlab's feasibility for deep learning.
  - Created framework for inference on final product.
  - Libraries used: TensorFlow, ClearML, tf2onnx, Deep Learning Toolbox (Matlab), ONNX Runtime.

Education

Academic Qualifications

- **Università degli Studi di Genova** — **Genova, Italy**
  Master's Degree in Robotics Engineering, 110/110 — *2021–Present*
  Robotics engineering is an interdisciplinary field of engineering that combines computer science, computer programming, artificial intelligence, and robotics to create, design, and manage robotic equipment and machinery.

- **Norwegian University of Science and Technology (NTNU)** — **Trondheim, Norway**
  Erasmus+ Exchange Student Program — *Aug. 2022– Dec. 2022*
  Opportunity to study in deep different subjects in the Cybernetics Department, such as Industrial Embedded Systems and Optical Remote Sensing.

- **Università degli Studi di Genova** — **Genoa, Italy**
  Bachelor Degree in Biomedical Engineering, 100/110 — *2018–2021*
  Information engineering is the engineering discipline that deals with the generation, distribution, analysis, and use of information, data, and knowledge in systems. The field first became identifiable in the early 21st century. Biomedical engineering takes part of this process and elaborates it for medical

- **Liceo Scientifico Luigi Lanfranconi** — **Genoa, Italy**
  High School Diploma, 81/100 — *2013–2018*
  The curriculum emphasises the link between the humanistic tradition and scientific culture. It covers a complete and widespread range of disciplines, including Italian language and literature, mathematics, physics, chemistry, biology, history, philosophy, Latin language and culture, English language and culture, art history and technical drawing.

- **Santa Cruz High School** — **Santa Cruz, California, USA**
  Exchange Student Program — *2016–2017*
  I've been living in Santa Cruz, California for almost one year of high school for an exchange student program to reach a proficiency English language level.

Notable Projects

- **Bachelor Thesis:** *'Development of an exergame for spatial orientation in Virtual Reality.'*
  I graduated in biomedical engineering with a thesis regarding the use of Virtual Reality in patients with neurological disorders, developing a video-game in Unity deepening spatial navigation which can represent an important element for diagnoses of these kinds of disorders. The project first asked some review of the state-of-the-art literature, then it involved the whole development of the code. I developed a frame where we could get important data from the game to process it later. Then, I developed some simple data analysis in Matlab.

- **Course Project:** *'TIAGo Follow Me!'*
  Assignment of SofAR class from Robotics Engineering course at Università degli Studi di Genova. The

assignment regards the development of a software architecture in ROS 2 where, in a simulation launched on Webots, TIAGo robot (made by Pal Robotics) follows a moving item seen by his RGB-D camera. To complete the task, we decided to have a different approach to the problem. Using libraries of computer vision and simple PID controllers for the robot, we implemented a reasonable architecture where TIAGo follows a target and tries to keep it at the center of its vision.

Technical and Personal skills

- **Programming Languages:** C, C++, C#, Python, PDDL, Arduino IDE, ROS, Matlab, Microchip Studio.
- **Industry Software Skills:** Most MS Office products.
- **General Soft Skills:** Good presentation skills, love teamwork, can write well organised and structured reports, always ready to listen and help those in need.
- **Language Skills:**
  - *Italian:* First Language.
  - *English:* Proficient level.
  - *Spanish:* Elementary level.
- **Other:** Organized schedule and routine.

Interests and extra-curricular activity

- I've always loved technology and everything related to PCs and modern machines. When I was 13, in the summer of my last year of middle school I built up my first PC with COTS at the time. With my PC, I just wanted to play video games but I accidentally started watching YouTube videos regarding the hardware world, which lead me to what I'm doing today.
- I'm deeply in love with reading books. During my high school, I've started a positive and beautiful collaboration with a professor who introduced me to the world of literature and philosophy. I usually read novels, journals and philosophy essays. I had the occasion to take part in philosophy competitions regarding modern problems and exhistential questions.
- I've been playing bass guitar for a long time, almost 8 years. I listen to a lot of different kinds of music appreciating every single particular they present. I own a collection of vinyls and in my spare time I create playlists in my Spotify account.
- I am also an avid hiker, strongly passionate about mountains and their surroundings.